

What is Statistics?

Statistics is the branch of mathematics used to **collect, analyze, interpret, visualize, and make decisions from data**. It helps turn raw data into insight — whether describing the data itself or drawing conclusions about a larger group based on a sample.

👉 Broadly, statistics has **two major fields**:

1. **Descriptive Statistics** – summarizing and describing the main features of a dataset.
2. **Inferential Statistics** – making predictions, estimations, and decisions about a population from sample data.

1. Population vs Sample

- 🧠 **Population**: The *entire group* you want to understand or draw conclusions about (e.g., all users of an app).
- 📊 **Sample**: A *subset* of the population selected for analysis.

Why samples?

Often it's *expensive or impossible* to collect data from everyone — so we use samples to estimate population characteristics (parameters).

Population parameter vs sample statistic

- Population measure → called a **parameter** (e.g., μ for mean, σ for standard deviation).
- Sample measure → called a **statistic** (e.g., $\bar{x}$ for mean, s for standard deviation).

2. Descriptive Statistics

Descriptive statistics describe datasets by summarizing data patterns without drawing conclusions beyond the data itself.

A. Measures of Central Tendency

These describe the “middle” of the data.

1. **Mean (Average)**

$$\bar{x} = \frac{\sum x}{n}$$

- Uses all data points.
- Sensitive to extreme values (outliers).

2. **Median**

- Middle value when data is ordered.
- Not influenced by outliers → better for skewed data.

3. **Mode**

- Most frequently occurring value.
- Useful for categorical data.

B. Measures of Dispersion (Spread)

Shows how much variability exists in the data.

1. **Range**

Range=Max-Min

- Simple but very sensitive to outliers.

2. **Interquartile Range (IQR)**

- Spread of the middle 50% of data:
IQR=Q3-Q1
- Good for skewed data.

3. **Variance (σ^2 / s^2)**

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n} \quad (\text{population})$$

- Measures average squared deviation from the mean.

4. Standard Deviation (σ / s)

$$SD = \sqrt{\text{variance}}$$

More interpretable than variance because it's in original units.

C. Frequency Distribution & Shape

Shows how often each value occurs.

Shape properties include:

- **Skewness** – measure of asymmetry of a distribution. Positive skew → tail on the right; Negative skew → tail on the left.
 - **Kurtosis** – measure of “tails” or extremity frequency. High kurtosis → heavy tails; low kurtosis → light tails.
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3. Probability Distributions

A **probability distribution** describes how values of a random variable are distributed.

A. Normal (Gaussian) Distribution

- Continuous, symmetric bell-curve.
- Defined by mean (μ) and standard deviation (σ).
- Most natural phenomena (like height, measurement errors) approx follow this.
- Formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

B. Binomial Distribution

Used for discrete random variables with **fixed number of independent trials** with two outcomes (success/failure).

- Formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- Mean = np and variance = np(1-p).

C. Poisson Distribution

Used for **count of events** in a fixed interval when events occur independently at a constant rate λ .

- Formula:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

- Mean = variance = λ .

◆ *Binomial → Poisson approximation*: When n is large and p small, binomial can be approximated by Poisson with $\lambda=np$.

4. Sampling & Probability Sampling Methods

Sampling techniques determine **how we select samples** from populations for analysis.

☑ Probability Sampling

Every member has a known chance of being selected.

1. **Simple Random Sampling**
Each individual equally likely to be chosen.
2. **Stratified Sampling**
Population divided into subgroups (strata); samples drawn from each.
3. **Cluster Sampling**
Population divided into clusters; clusters randomly selected, then all or some elements sampled.

5. Central Limit Theorem (CLT)

One of the **most fundamental results in inferential statistics**.

Definition:

If you take *many random samples* of size n from any population (regardless of its shape), the **distribution of the sample means** will approach a **Normal distribution** as n becomes large (typically $n \geq 30$).

Key Points:

- Sampling distribution of the mean $\rightarrow$ Normal.
- Mean of sampling distribution = population mean μ .
- Standard deviation of sampling distribution (called *standard error*) = $\sigma / \sqrt{n}$.

CLT is crucial because it justifies using normal-based inference (confidence intervals, hypothesis tests) even when the original distribution is unknown.

6. Inferential Statistics

Unlike descriptive stats (summary), **inferential statistics** help you **make predictions or generalizations** about a population from a sample.

Common techniques include:

- **Confidence intervals**
- **Hypothesis testing (t-test, χ^2 test, ANOVA)**
- **Regression and correlation analysis**

All rely heavily on understanding sampling distributions and CLT.

Summary: Black-Box to Statistics

Topic	Key Concept
Descriptive Stats	Summarize data: mean, median, mode, range, variance, SD
Distribution	How values/ probabilities spread; Normal/Binomial/Poisson

Topic	Key Concept
Sampling	How to pick representative data (random, stratified, cluster)
CLT	Sample means $\rightarrow$ Normal distribution
Inferential Stats	Test hypotheses, estimate population metrics
